# SE LAB 6 — Coverage Analysis Report

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| **Application** | Phone Repair Quote |
| **Test Framework** | pytest |
| **Coverage Tool** | coverage.py |

## 1. Objective

The objective of this lab is to test the Phone Repair Quote application and measure both code (line) coverage and branch coverage. The tests were written using pytest and coverage.py, with the goal of achieving at least 90% coverage.

## 2. Code Coverage Analysis

The completed code-coverage test suite executed 20 tests successfully. The final coverage report showed 40 statements in phone_repair_quote.py, with 0 missed statements and 100% code coverage.

| File | Statements | Missed | Coverage |
|---|---|---|---|
| phone_repair_quote.py | 40 | 0 | 100% |
| test_phone_repair_quote.py | 46 | 0 | 100% |
| Total | 86 | 0 | 100% |

## 3. Branch Coverage Analysis

A separate branch-coverage test suite was created to exercise both outcomes of conditional logic in the application. The combined branch-coverage run executed 44 tests successfully. The final report showed 28 branches, with 0 partial branches and 100% branch coverage for phone_repair_quote.py.

| File | Statements | Branches | Partial | Coverage |
|---|---|---|---|---|
| BranchCoverageTest_phone_repair_quote.py | 54 | 0 | 0 | 100% |
| phone_repair_quote.py | 40 | 28 | 0 | 100% |
| test_phone_repair_quote.py | 46 | 0 | 0 | 100% |
| Total | 140 | 28 | 0 | 100% |

## 4. Test Coverage Summary

| Metric | Result | Requirement | Status |
|---|---|---|---|
| Code / Line Coverage | 100% | ≥ 90% | PASS |
| Branch Coverage | 100% | ≥ 90% | PASS |
| Branch Test Execution | 24 passed | All tests pass | PASS |
| Combined Coverage Execution | 44 passed | All tests pass | PASS |

## 5. Coverage Test Design

The tests cover the major decision paths of the application, including valid and invalid repair parts, budget and flagship brand tiers, screen and non-screen labour paths, warranty and non-warranty conditions, express and normal repairs, technician availability, parts availability, trade-in age ranges, invalid inputs, and cracked versus non-cracked screens.

## 6. Conclusion

The Phone Repair Quote application achieved 100% code coverage and 100% branch coverage. All executed tests passed successfully. Therefore, the coverage results exceed the lab requirement of at least 90% for both code and branch coverage.

Note: The terminal screenshots showing the code-coverage and branch-coverage results can be added to this report before final GitHub submission.